

Complete Project Summary

Introduction, problem, objectives, methodology, implementation, architecture, tech stack, flow, tests, limitations, and conclusion.

Source brief for a detailed report, PPT, or viva.

16 September 2026 | Version 0.1.0 | Final-year college project

1. Introduction

BhoomiScan (tagline: Your Alvocate.) is an AI-assisted residential property due-diligence web application. A logged-in user creates a residential property file, uploads PDFs or images, runs Sarvam Document AI for OCR and Kannada-to-English translation, then a deterministic Python risk engine (not an LLM) scores five risk categories, compares multiple documents, builds a versioned report, SHA-256 hashes it, stores a local blockchain-style hash chain, and verifies integrity.

It helps a buyer check papers before buying. It is not a lawyer, not a government portal, not a payment product, and not a legal guarantee. Focus: Karnataka-style papers, including Kannada scans. UI: dark navy, gold, ivory type, rounded royal cards, simple English.

Product	BhoomiScan - Your Alvocate.
Type	Final-year college project
Version	0.1.0
Frontend	http://localhost:5173
Backend	http://127.0.0.1:8000
Health	GET /health
Workspace	BhoomiScanV18

2. Problem statement

- Property papers are scanned, often in Kannada, and hard for a buyer to read.
- Risk sits in five areas: ownership/title, litigation, mortgage/encumbrance, approval/construction, property record/boundary.
- Buyers often treat missing papers as no problem. That is dangerous.
- Cross-document mismatches (Sale Deed survey 45/3 vs Survey Sketch 45/4) are easy to miss.
- There was no simple local student tool that does OCR, translate, extract, evidence, risk, recommend more papers, combine the case, report, and tamper-evident hash.
- Commercial tools may be costly, English-only, or overclaim that a property is legally clear.

Academic question: Can a student-scale full-stack system use Indic Document AI plus deterministic rules to show DETECTED / NOT_VERIFIED / NO_ISSUE_FOUND with evidence, without faking AI, payments, or a public blockchain?

3. Objectives

3.1 Primary

Build an end-to-end local web system: Upload -> AI analysis -> more documents -> compare -> recalculate risks -> coverage and risk score -> final report -> SHA-256 -> local ledger -> verify.

3.2 Specific

- JWT auth (username + password, bcrypt).
- Property case per user; only RESIDENTIAL analysis is live.
- Upload PDF/PNG/JPG, max 10 MB, magic-byte check, disk + PostgreSQL metadata.
- Sarvam digitise, Kannada translate in 1800-char chunks, optional structured extract.
- Classify document type (sale deed, EC, khata, and others).
- Five risk categories; statuses DETECTED / NO_ISSUE_FOUND / NOT_VERIFIED; levels HIGH / MEDIUM / LOW / INFO.
- Never treat missing evidence as a clean title.
- Recommend missing documents; multi-document vault; delete own files.
- Deterministic comparison (no LLM for MATCH/MISMATCH).
- Recalculate from stored OCR/translation/extraction; do not re-call Sarvam unless Analyze is clicked.
- Verification coverage percent is not the same as risk score.
- Versioned report PDF + JSON; new version if regenerated; never silently edit a finalized report.
- SHA-256 of canonical JSON; local GENESIS hash chain; integrity VALID or TAMPERED.
- 401/403/404 ownership; pytest with Sarvam mocked.

3.3 Not implemented (do not invent these)

- Public blockchain, Bitcoin, Ethereum, mining, wallets, smart contracts, cryptocurrency.
- Real payments / UPI / cards. Subscription page is UI only (Get Subscription).
- Real Google / X / phone OAuth (demo buttons only).
- Government portal scraping (Kaveri, Bhoomi, BBMP, etc.).
- Custom ML training, microservices, Kubernetes.
- Agricultural / Commercial / Industrial analysis (names kept; gated to subscription UI).
- Any claim that the property is legally clear or guaranteed safe.

4. Scope by phase

Phase 1 - Foundation

Landing, signup, login, JWT, dashboard, domain select, residential types, property case, document upload, protected routes, dark royal UI. Agricultural / Commercial / Industrial show Get Subscription (not Coming Soon and not homes-only). 13 tests in tests/test_phase1.py.

Phase 2 - Real AI document analysis

Sarvam digitise + translate + extract; classifier; merge extraction with source-text veto (AI cannot hide a stay/case if the text has it); evidence snippets; Python risk engine; recommendations; analysis UI. 7 tests in tests/test_phase2.py. Sarvam mocked in pytest.

Phase 3 - Case-level due diligence

Vault, multi-doc, delete, recalculate, MATCH/MISMATCH/NOT_AVAILABLE, coverage, risk score, finalize dialog, versioned reports, SHA-256, ledger, verify, PDF. tests/test_phase3.py. Last known combined pytest: 36 passed.

5. Methodology

5.1 Hybrid AI + rules

Sarvam Document AI reads, OCRs, translates, and optionally extracts JSON. Python classifies (fallback), merges fields, decides risk, compares documents, scores, hashes, and writes the ledger. An LLM does not decide risk or MATCH/MISMATCH.

5.2 Three-state verification (critical)

DETECTED = explicit adverse evidence (stay, mismatch, incomplete names, unauthorized construction, uncleared mortgage). NO_ISSUE_FOUND = a relevant document explicitly clears it (example: EC says nil encumbrance). NOT_VERIFIED = not enough supporting papers. This is the default for absence. Wrong: No EC means no mortgage. Right: No EC means Mortgage NOT_VERIFIED.

5.3 Evidence-first

Findings cite source snippets. If seller/buyer names are missing but survey 45/3 exists, evidence still shows the survey snippet and notes that names were not found.

5.4 Case vs document

Document analysis is one file and may call Sarvam. Case recalculate uses all analyzed files, reuses stored text/JSON, merges risks, compares, and scores.

5.5 Coverage vs risk

Risk score: points only from DETECTED findings. HIGH=20, MEDIUM=10, LOW=5, cap 100. NOT_VERIFIED adds 0. Coverage: Verified=1.0 (DETECTED or NO_ISSUE_FOUND), Partial=0.5 (NOT_VERIFIED but a related paper was uploaded), Not Verified=0.0. Percent = $\text{round}(100 * \text{sum} / 5)$. Coverage is not a safety guarantee.

5.6 Tamper evidence

Canonical JSON -> SHA-256 report_hash. Ledger hashes report_id, report_hash, previous_hash, timestamp. First previous_hash is GENESIS. Verify recomputes hashes; it does not blindly trust the stored hash. Chain is checked by report version, not timestamps, to avoid timezone false tamper.

5.7 Testing

FastAPI TestClient, SQLite in-memory, StaticPool, mocked Sarvam. Live UI uses SARVAM_API_KEY in the project-root .env.

6. Tech stack

Backend

- Python 3.11+ (development also ran 3.14.7).
- FastAPI, Uvicorn, SQLAlchemy 2 with UUID primary keys.
- PostgreSQL 16 via Docker Compose: database/user/password bhoomiscan, port 5432.
- Tests use SQLite in-memory.
- Pydantic v2, pydantic-settings, PyJWT HS256, bcrypt.
- python-multipart uploads; sarvamai SDK (digitise, status, results, download, translate).
- requests/httpx; fpdf2 PDFs; pytest.

Frontend

- React 19, Vite 6, React Router 7, Tailwind 3, Framer Motion, Axios.
- localStorage key bhoomiscan.token.
- VITE_API_URL example http://localhost:8000. Analyze timeout 180 seconds.

Environment

Root .env: DATABASE_URL, JWT_SECRET, JWT_ALGORITHM, JWT_EXPIRE_MINUTES (1440), MAX_FILE_SIZE_MB (10), CORS_ORIGINS, SARVAM_API_KEY, poll 3s, max wait 120s. Never expose API key, JWT secret, password hashes, or filesystem paths to the frontend. Files live at storage/documents/{case_id}/. Single FastAPI process and single Vite app. No extra microservices.

7. Architecture

Browser SPA talks to FastAPI with Bearer JWT. Routers: auth, property cases and uploads, per-document analyze, recalculate/compare, finalize/hash/ledger/verify/PDF. PostgreSQL holds metadata, OCR text, extracted JSON, analyses, comparisons, reports, ledger. Disk holds original files. Sarvam is called only on Analyze, not on Recalculate.

Layers: (1) UI royal theme and protected routes. (2) REST with ownership checks. (3) Domain services. (4) SQLAlchemy models. (5) Sarvam I/O. Split AI reading from deterministic rules so results are explainable and testable.

8. User flow

Landing -> Signup (username, password, confirm) -> Login (JWT) -> Dashboard/Home (Start Property Analysis + upgrade cards; no fake stats) -> Property domain. RESIDENTIAL continues. AGRICULTURAL / COMMERCIAL / INDUSTRIAL go to /subscription (Get Subscription). Residential types: vacant land, independent house, apartment/flat, villa, plot. Create property case. Document vault: upload, Analyze, recommended papers, Recalculate. Due diligence: overall risk, score /100, coverage %, five cards, evidence, MATCH/MISMATCH/NOT_AVAILABLE, unverified CTAs. Finalize dialog: Some checks may remain unverified. Missing documents do not block the report. Report: VALID/TAMPERED, SHA-256, record number, timestamp, Verify Integrity, Download PDF.

Main demo (Kannada litigation sale deed)

- Upload SaleDeed_01_Litigation_Kannada.pdf and Analyze. May classify as SALE_DEED or COURT_CASE_DOCUMENT if O.S. No. dominates keywords. Litigation DETECTED/HIGH (stay, pending O.S. 234/2019, transfer restriction). Mortgage and approval NOT_VERIFIED.
- Upload EC with explicit nil encumbrance, Analyze, Recalculate. Mortgage may become NO_ISSUE_FOUND.
- Upload Survey Sketch. Equal survey MATCH; different survey MISMATCH HIGH.
- Upload Building Approval with explicit approval. Approval may become NO_ISSUE_FOUND.
- Recalculate -> Finalize -> Verify VALID.

9. Data model

- User: id, unique username, password_hash, timestamps.
- PropertyCase: user_id, domain, property_type, status ACTIVE.
- Document: filenames, file_path, mime, size, status UPLOADED/PROCESSING/ANALYZED/FAILED, document_type, confidence, language, original_text, translated_text, extracted_data JSON, processing_error.
- Analysis: case_id, document_id, COMPLETE/FAILED/PROCESSING, pipeline_steps, extracted_data.
- RiskFinding: category, status, risk_level, summary, finding_data JSON.
- EvidenceItem: document_id, type, page, source_text, extracted_field, confidence.
- Recommendation: category, document_type, reason.
- DocumentComparison: field, document A/B, values, MATCH/MISMATCH/NOT_AVAILABLE, severity, explanation.
- FinalReport: version, report_content JSON, report_hash, risk_score, risk_level, verification_coverage. Not

silently edited; regenerate = new version.

- LedgerRecord: report_id unique, report_hash, previous_hash, record_hash, hashed_at ISO UTC.

Startup: create_all plus ensure_schema adds Phase 2 columns on existing documents tables without dropping data.

10. API endpoints

JWT required except signup, login, and health. Case/document/report routes check ownership (401/403/404).

POST /auth/signup	201; 409 duplicate; 422 mismatch
POST /auth/login	Bearer access_token
GET /auth/me	Current user
POST /property-cases	Residential only; else 400 subscription
GET /property-cases	Own cases
GET /property-cases/{id}	403/404
POST .../{id}/documents	Upload
GET .../{id}/documents	List
DELETE .../documents/{doc}	Owner; delete file
POST .../documents/{doc}/analyze	Sarvam; long timeout
GET .../documents/{doc}/analysis	Latest document analysis
GET .../{id}/analysis	Latest analysis on the case
GET .../{id}/recommendations	From latest analysis
GET .../{id}/evidence/{fid}	Finding evidence
POST .../{id}/recalculate	All analyzed documents
GET .../{id}/due-diligence	Same as recalculate
GET .../{id}/comparisons	Stored comparisons
POST .../{id}/finalize	New report + ledger
GET .../{id}/reports	Version list
GET .../reports/{rid}	One report
GET .../reports/{rid}/verify	VALID or TAMPERED
GET .../reports/{rid}/pdf	Download PDF
GET /health	Public

The frontend must not send risk scores for the backend to trust. The backend always computes them.

11. Frontend routes

Public: / Landing, /login, /signup. Protected: /dashboard Home, /ai, /profile, /profile/settings, /contact, /subscription, /property-domain, /property-type, /property-case/:id/upload vault, /property-case/:caseId/documents/:documentId/analysis, /property-case/:id/due-diligence, /property-case/:id/reports/:reportId. Logged-in nav: Home, AI, profile, Sign Out. Stepper: Property kind, Home type, Document vault, Due diligence.

12. Implementation algorithms

12.1 Upload

Check extension, declared MIME, and magic bytes (%PDF, PNG, JPEG). Reject empty/too large files. Sanitize names. Never execute uploads.

12.2 Analyze pipeline

- Status PROCESSING.
- Sarvam digitise (Kannada); poll; read get_results page blocks; ZIP download fallback; strip NUL before PostgreSQL TEXT.
- If Kannada script, translate in 1800-character chunks to English.
- Keyword classifier on filename+text; confidence below 0.45 -> OTHER.
- Sarvam extract by schema; on failure use empty object.
- merge_extraction: regex fallback + AI; source veto for stay/case/restriction.
- evaluate_risks + build_recommendations; persist; ANALYZED or FAILED.

Extracted groups: property (survey, khata, area, village, hobli, taluk, district, state, type); ownership (seller, buyer, owner); transaction (sale amount, registration, previous deed); litigation; mortgage; approval; boundaries N/S/E/W.

12.3 Risk engine

- Ownership: missing seller/buyer -> DETECTED MEDIUM; identifiers without title-support docs -> NOT_VERIFIED; never auto-clear title.
- Litigation: pending/stay/restriction/case number/source hits -> DETECTED; explicit no pending litigation on verify-types -> NO_ISSUE_FOUND; else NOT_VERIFIED. Word-boundary lien so alienation is not a mortgage.
- Mortgage: charge without closure -> DETECTED; EC + nil/no/free from encumbrance -> NO_ISSUE_FOUND; else NOT_VERIFIED.
- Approval: unauthorized/deviation -> DETECTED; approval-type docs + present/OC -> NO_ISSUE_FOUND; a sale deed alone is not permission.
- Property record: identifiers still NOT_VERIFIED until cross-doc; mismatches on recalculate -> DETECTED.
- Case merge: DETECTED wins; else any NO_ISSUE_FOUND; else NOT_VERIFIED; then overlay MISMATCH.

12.4 Comparison

Compare seller, buyer, owner, survey, khata, area, village, hobli, taluk, district, property type, four boundaries, registration number/date, previous deed number. Normalize whitespace, punctuation, prefixes (Survey No., Sy. No.). Names: collapse space and casefold; do not merge different names. Area to sq ft (sq m * 10.7639), 2% tolerance. Fewer than two documents with a value -> NOT_AVAILABLE. Else pairwise MATCH or MISMATCH with severity and explanation.

12.5 Overall risk

HIGH if any DETECTED HIGH or score >= 40. MEDIUM if any DETECTED MEDIUM or score >= 10. Else LOW.

12.6 Report, hash, ledger

report_content includes property info, documents, risk summary, overall, coverage, detected issues, evidence, mismatches, unverified, recommendations, disclaimer. report_hash = SHA-256(canonical JSON with sorted keys). record_hash = SHA-256 of report_id, report_hash, previous_hash, timestamp. Verify: content hash matches; ledger material hash matches; previous is GENESIS or prior version record_hash.

12.7 Recommendations

If a category is DETECTED or NOT_VERIFIED, suggest missing types not already uploaded: parent deed, EC, khata, mutation, court order, bank NOC, building approval, survey sketch, and related papers.

13. Legal language

Allowed: Potential issue detected. Based on the documents provided. Could not be verified. Further verification is recommended. AI-assisted document review. Forbidden: Property is completely safe / legally clear / guaranteed clean. Do not say no litigation exists unless a proper document explicitly says so (then NO_ISSUE_FOUND, still not a court guarantee).

Disclaimer stored in reports: This report is an AI-assisted document review and is based only on the documents provided. It does not replace advice from a qualified lawyer, surveyor, government authority, or other professional.

14. Security

- Passwords hashed with bcrypt; never returned.
- JWT Authorization header; 401 without token; 403 other user's case.
- Non-residential case creation is gated; not a real payment.
- Upload type sniffing; strip NUL from OCR; CORS limited to Vite origins.
- Secrets only in server .env.

15. Tests

Phase 1	test_phase1.py - signup/login/JWT, residential case, commercial 400, PDF upload, 403/404, re
Phase 2	test_phase2.py - analyze auth, mocked Kannada deed, 403/404, litigation evidence, absence is
Phase 3	test_phase3.py - multi-doc, MATCH/MISMATCH, recalculate, EC clears mortgage, coverage, score
Frontend	npm run build (Vite)
Last known run	36 passed

16. Known limitations

- Live OCR depends on Sarvam and scan quality.
- A litigation-heavy sale deed may classify as Court Case Document.
- Boundary OCR lines can concatenate (north absorbing south/east/west).
- Kannada party names may not extract; ownership DETECTED incomplete by design, not clear.
- Ledger is a local hash chain, not Ethereum.
- Subscription / Google / X / phone are UI demos. No live government fetch.
- Recalculate does not re-OCR. Local student deployment, not production-hardened.

17. Key source files

Backend: app/main.py, config.py, constants.py, database.py, db_migrate.py, deps.py. Routers: auth, property_cases, analysis, reports. Models: user, property_case, document, analysis, report. Services: auth, storage, sarvam_service, document_classifier, extraction_service, evidence_service, risk_engine, recommendation_engine, analysis_service, comparison_service, scoring_service, case_analysis_service, hashing, ledger_service, report_service, report_pdf.

Frontend pages: Landing, Login, Signup, Home, PropertyDomain, PropertyType, DocumentUpload (vault), DocumentAnalysis, DueDiligence, FinalReport, Subscription, Profile, AI, Contact. Also docs/Phase1_Test_Case_Report.pdf and Phase2_Test_Case_Report.pdf.

18. Conclusion

BhoomiScan is a complete local, explainable pipeline for residential due diligence: Indic Document AI to read Kannada papers, Python rules for risk and comparison, and a tamper-evident report without fake payments or a public blockchain.

Academic points: (1) three-state verification so missing papers stay NOT_VERIFIED; (2) hybrid AI plus rules; (3) case-level comparison and coverage versus risk; (4) SHA-256 plus GENESIS hash chain with recomputed integrity. A buyer can go from signup to a versioned PDF with evidence and VALID/TAMPERED verification, while the system refuses to call an unverified property clear.

19. Future work (not built)

- Better party-name NER and user-confirmed document type.
- Government API integration (optional, if legally allowed).
- Real payments if a product is needed.
- Optional RFC 3161 timestamps; mobile app.

20. How to expand this PDF into a full report

Another writer or AI can turn this into a 15-25 page report, a 12-15 slide PPT, or viva Q&A. Include introduction, problem, objectives, related work (manual due diligence vs this tool), methodology, architecture diagrams, module-wise implementation, algorithms, ER-style model, API table, screens (landing, vault, analysis, due diligence, VALID report), tests, limitations, future work, conclusion, and disclaimer. Do not invent live blockchain, live UPI, live OAuth, agricultural analysis, or legal certainty. Do not omit the NOT_VERIFIED rule, five categories, Sarvam vs Python split, GENESIS ledger, 36 tests, or Get Subscription gating for Agricultural, Commercial, and Industrial.